Est. SEP 9

miércoles, septiembre 09, 2026 1:51 PM

$A = B \rightsquigarrow$ ⓘ $A \subseteq B$
(ii) $B \subseteq A$.

El conjunto Vacio : $\phi = \{\}$

Propo: $\phi \subseteq A \;\forall A$ conjunto.

Prueba: Sea $x \in \phi \longrightarrow x \in A$

$P \longrightarrow q$ QED

Propo: El vacio es único.

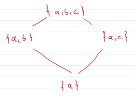
Dem. Suponga que $\phi_1 \wedge \phi_2$ son dos conjuntos vacios.

$$\phi_1 \subseteq \phi_2 \wedge \phi_2 \subseteq \phi_1 \Rightarrow \phi_1 = \phi_2 \quad QED$$

Relación $\leq$ C.O.P.O = Conjunto Parcialmente Ordenad.

Reticulo.

(i) Reflexividad $A \subseteq A$.
(ii) Antisimetría $A \subseteq B \wedge B \subseteq A \Rightarrow A = B$ (Extensionalidad)
(iii) Transitividad $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$.



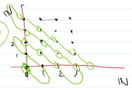
Conjunto Potencia.

infinitos $\longrightarrow$ $\mathbb{N} = \{0, 1, 2, 3, 4, 5, \dots\}$

$\mathbb{Z} = \{\dots -3, -2, -1, 0, 1, 2, 3, 4 \dots\}$

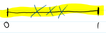
$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$

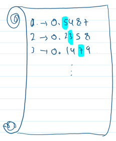
$\mathbb{N} \times \mathbb{N} \Rightarrow$



$\mathbb{Q}$

$\mathbb{R} \to (0,1)$





0. → 0.5487
2 → 0.2358
3 → 0.1479
⋮

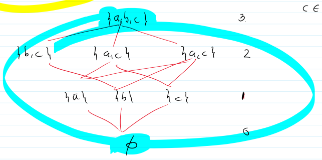
0.648

$$|\mathbb{N}| < |\mathbb{R}|$$

$\wp(\mathbb{N}) \rightarrow$ Conjunto potencia.

$\{a,b,c\} = A \qquad \wp(A) = \{B : B \subseteq A\}$

$c \in A$



$$\wp(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

$$|\wp(A)| = 2^{|A|} = 2^3 = 8.$$

$$|\mathbb{N}| < |P(\mathbb{N})| < |\wp(\wp(\mathbb{N}))|$$

Teorema: Sea A un conjunto:
prueba.

(i) $\phi \in \wp(A)$. Porque $\phi \subseteq A$.
(ii) $A \in \wp(A)$. Porque $A \subseteq A$
(iii) $\wp(A) \neq \phi$. Porque $\phi \in P(A)$.

$A = \phi \qquad |A| = 0$
$\wp(A) = \{\phi\} \qquad |\wp(A)| = 1 = 2^0$

(iv) $A \subseteq B \Rightarrow \wp(A) \subseteq \wp(B)$.

Dem: Hipótesis: $A \subseteq B$.

Mostraré : $\wp(A) \subseteq \wp(B)$.

Sea $X \in \wp(A) \Rightarrow X \subseteq A$ por hip. $A \subseteq B$.
por transitivida $X \subseteq B \Rightarrow X \in \wp(B)$.

(v) $A = B \iff \wp(A) = \wp(B)$.

Dem: $\Rightarrow$ Hipótesis: $A = B$. Mostrare que $\wp(A) = P(B)$

por (iv) $A \subseteq B \Rightarrow P(A) \subseteq P(B)$
$B \subseteq A \overset{(iv)}{\Rightarrow} \wp(B) \subseteq P(A)$ } ext $\Rightarrow P(A) = P(B)$

$\Leftarrow$ Hipótesis $\wp(A) = P(B)$ Mostraré : $A = B$.

$\subseteq$ Sea $x \in A \Rightarrow \{x\} \subseteq A \Rightarrow \{x\} \in P(A) \Rightarrow$
$\{x\} \in P(B) \Rightarrow \{x\} \subseteq B \Rightarrow x \in B$.

$\supseteq$ Sea $y \in B \Rightarrow \{y\} \subseteq B \Rightarrow \{y\} \in P(B) \Rightarrow$
$\{y\} \in P(A) \Rightarrow \{y\} \subseteq A \Rightarrow y \in A$.

Sin perdida de generalidad $B \subseteq B$.

$A = \{1\} \quad B = \{1,2\} \quad C = \{\overset{A}{\{1\}}, \{2\}\} \quad D = \{1, \{1\}, 2, \{2\}\}$

$A \in C$ ✓ ~~$A \subseteq C$~~ $A \in D$ ✓ $A \subseteq D$ ✓

$X = \{1,2\} \qquad P(X) = \{\phi, \{1\}, \{2\}, \{1,2\}\}$